

S2 Supplementary Data

Audited validation summaries and explicit unavailable result families

Short title: S2 Data - NeuralManifoldDynamics

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S2.0 Global coordinate contrast family

An article-level global `mnp3_3d/coords_9d` pairwise contrast family with a single BH-FDR correction is **not available**. The manuscript is a measurement contract paper, not a pre-registered Task A versus Task B coordinate experiment. The original template's global-null table is therefore intentionally not filled. Dataset-specific contrasts below must not be read as a replacement for that missing article-wide family.

S2.1 ds003838 primary validation contrasts

The full ds003838 export completed for 130 subjects, 6,543 blocks, and 27,670 block-native windows with a source-window match fraction of 1.0. The corrected stage-statistics run used 26,460 windows and $N = 62$ subjects with complete stage coverage. The table reports values from

project/diary/127_20260625_full_stats_package_ds003838.md; the two window counts belong to different run directories and are not pooled.

Metric	χ^2	Friedman p	Interpretation
traj_path_length	226.2	8.8×10^{-48}	stage-dependent
traj_efficiency	221.8	7.8×10^{-47}	stage-dependent
vagal_index	88.0	3.4×10^{-18}	stage-dependent
sympathetic_index	63.7	4.8×10^{-13}	stage-dependent
ecg_hrv_hr_mean_bpm	62.6	8.3×10^{-13}	stage-dependent
ecg_hrv_sdnns_ms	62.0	1.1×10^{-12}	stage-dependent
ecg_hrv_rmssd_ms	24.1	7.7×10^{-5}	stage-dependent
anchor_index	13.6	8.6×10^{-3}	stage-dependent
traj_mean_curvature	7.6	0.11	not significant

The registered listen–memory contrasts for vagal_index were:

Contrast	Cohen's d	Wilcoxon p
listen vs mem5	+1.718	6.9×10^{-11}
listen vs mem9	+1.750	1.8×10^{-10}
listen vs mem13	+1.995	1.2×10^{-10}

The later one-line summary in the same diary contains a different +2.085 value for a row labelled C3. That contrast definition is not resolved here; the explicit pairwise table above is the value used for the manuscript.

S2.2 Regional contrasts

The contract supports regional fMRI network and EEG channel-group trajectories, regional Jacobians, and optional regional stratified outputs. The cited ds003838 run enables regional trajectories but not regional block Jacobians. No complete article-level table of regional coordinate contrasts with a predeclared multiple-comparison family is archived in the current evidence pack. Regional capability is therefore reported as implemented, while a full regional inferential table remains unavailable.

S2.3 Block-level contrasts

The release-bound ds003838 block-native aggregate is descriptive coverage evidence:

Stage	Blocks	Windows
rest	65	3,961
listen	2,171	7,944
mem5	1,273	2,905
mem9	1,469	5,361
mem13	1,565	7,499
Total	6,543	27,670

These totals establish a successful block/window join for the full export. They are not the inferential sample for the corrected stage-statistics package.

S2.4 Secondary coupling

The ds003838 statistics package reports all 12 inter-network `coupl_*` terms as stage-dependent, with Friedman statistics approximately 65–97 and uncorrected **p** values from 2.6×10^{-13} to 5.4×10^{-20} . The archived summary identifies central-to-parietal and temporal-from-central coupling as among the strongest effects.

These are stage-wise coupling results, not an EAP partial-correlation table. An article-level table of partial correlations, covariate choices, and FDR family size is not locked and remains unavailable.

S2.5 Secondary physiology–coordinate coupling

HRV and anchor indices are exported on the same block-native grid and are included in the ds003838 stage statistics. This establishes a usable physiology–coordinate measurement surface and stage dependence under the specified task contract. It does not establish causality, a general autonomic biomarker, or a cross-dataset coupling law.

S2.6 Physiological and temporal confounds

No single combined partial-correlation table exists across all follow-up scripts. The auditable confound results are instead distributed across QC artifacts:

- ds006848 ECG polarity was inverted in 92.7% of feature rows before correction; the corrected median HR was 76.2 bpm and RMSSD 40.7 ms.
- 87.7% of 60 s ds006848 working-memory HRV windows overlapped retrieval events, so working-memory HRV inference was withheld.
- The ds006848 frontal 30–45 Hz artifact contrast differed by condition but pointed opposite to the manifold m/d ordering; temporal muscle power was not significant ($p = 0.61$).
- Window overlap and event-purity results are reported in the 04c/04d scripts, while LOOSO/ bootstrap results are in 04e. These are cohort-specific audits, not one combined FDR family.

S2.7 Descriptive summary by task

The available task descriptors are the ds003838 stage labels rest, listen, mem5, mem9, and mem13, together with the block/window counts in S2.3. The reference ds004511 metadata snapshot contains 132 HDF5 outputs spanning 45 subjects and three tasks. Tonic SCL, EMG-RMS, and other descriptive columns are not inserted here unless a locked run table supplies their exact aggregation and source path.

S2.8 Analysis-owned synthetic validation

Two analysis-owned synthetic studies provide bounded numerical validation of the existing local-dynamics candidates. They are not additional empirical datasets and do not alter the ingest measurement contract. The prospective protocols, frozen numeric specifications,

and outcome reports are retained in `project/astra/m3_chart_20260906/` and `project/astra/m3_extension_20260906/`.

Study	Fixed design	Computed evidence	Status and scope
M3 benchmark	27 scenario cells; 216 comparator rows	Known discrete 3D truth; existing local mean and affine candidates plus controls	PASS for implementation; PARTIAL PASS for M3 qualification
CHART	96 points; fixed mixed rotation, isotropic and anisotropic maps	Affine transformed-vector discrepancy at most 2.9×10^{-14} ; covariance checked through a transformed Cholesky factor	PASS for implementation and bounded coordinate checks
M3 extension	6 cases; 12 training fits; 3 held-out paths $\times 48 = 144$ fixed queries per case	288 fitted comparator rows with finite predictions; six oracle rows reported separately	PASS for bounded implementation and frozen synthetic experiment; PARTIAL PASS / METHOD-LIMITED for scientific qualification

The M3 extension compares local-mean and local-affine candidates at the same held-out queries.

The values below are means over the twelve training fits; normalized RMSE is divided by the case-specific truth RMS, and alignment rank is Spearman correlation with the same-neighbour oracle alignment.

Synthetic case	Mean nRMSE	Linear nRMSE	Mean rank	Linear rank
full linear 3D	0.338	0.369	0.938	0.946
full nonlinear 3D	0.330	0.349	0.963	0.961
full linear 9D	0.536	0.878	0.787	0.772
full nonlinear 9D	0.554	0.768	0.809	0.770
latent linear 9D $\rightarrow$ observed 3D	0.347	0.391	0.889	0.916
linear 3D + observation noise	0.295	0.279	0.949	0.959

The two cross-segment order-corruption controls had normalized RMSE values of 1.152–1.720 and alignment ranks from -0.096 to $+0.296$. They use a global circular source-order roll across held-out segments; they are an artificial control and not a within-recording temporal null or an independent null sample. The full linear and nonlinear 9D simulations are known-truth test systems,

not the canonical neural coords_9d feature object. The exact target in the partial and noisy cases is conditional on the current observed state, not latent velocity or a history-conditioned forecast.

The Gaussian memory witness is approximately 6.58×10^{-9} in Frobenius norm for the partial-observation case and 0.0441 for the noisy case. Fully observed linear 3D and 9D witnesses are approximately 2.0×10^{-17} and 4.4×10^{-17} , respectively. The small partial-observation value is a resolved but tiny model diagnostic; the noisy value shows that current-observation target recovery does not by itself establish observed-state sufficiency. These values are not effect sizes, biological evidence, or acceptance thresholds.

The reports retain the source, runtime, protocol, numeric-specification, and artifact hashes.

The compact provenance map is:

Study	Fixed design	Frozen references	Evidence/status
M3 bench-mark	27 scenario cells; 216 comparator rows	m3_numeric_spec.json (SHA256 prefix 0e7cec90da5e)	PASS for implementation and bounded affine checks; PARTIAL PASS for M3 qualification
CHART	96 points; fixed mixed rotation, isotropic and anisotropic maps	chart_numeric_spec.json (SHA256 prefix 36adcfb98985); chart_final.json	Affine transformed-vector discrepancy at most 2.9×10^{-14} ; covariance checked through a transformed Cholesky factor
M3 extension	6 cases; 12 training fits; 3 held-out paths $\times$ 48 = 144 fixed queries per case	numeric_spec.json; freeze_manifest.json; primary artifact	288 fitted comparator rows with finite predictions; six oracle rows reported separately; PARTIAL PASS / METHOD-LIMITED for scientific qualification

Full SHA256 values and runtime details remain in the cited report.md files. Candidate status remains candidate_not_qualified; no biological claim or automatic scientific PASS is emitted.

Bibliography

Competing Interests

The authors have declared that no competing interests exist.